Relations and Functions

Concept

A function gives exactly one output for each allowed input.

Domain / range

Allowed inputs / resulting outputs.

Vertical-line test

A function has at most one intersection with any vertical line.

Notation

$f(4)$ asks for an output; $f(x) = 4$ asks which inputs give 4.

Context

Use discrete sets for counts; intervals for continuous inputs.

Example 1

Is $\{(1,4), (2,4), (3,7)\}$ a function? State its domain and range.

Solution

1. Yes: each input has only one output. Repeated outputs are allowed.

2. Domain $\{1,2,3\}$; range $\{4,7\}$.

Relations and Functions

Example 2

For $f(x) = 3x - 4$, find $f(-2)$ and solve $f(x) = 8$.

Solution

1. Evaluate: $f(-2) = 3(-2) - 4 = -10$.

2. Solve: $3x - 4 = 8 \rightarrow x = 4$.

Relations and Functions

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Is $x = y^2$ a function of x over the full real relation? Explain.

2. For $g(x) = -2x + 5$, find $g(3)$ and solve $g(x) = 9$.

3. State domain and range for $\{(0,-1), (2,3), (4,3)\}$.

4. Explain why a continuous line may misrepresent a ticket-sales model.

Relations and Functions

Practice answers | check after trying

1. No. At $x = 4$, both $y = 2$ and $y = -2$ occur.

2. $g(3) = -1$; $-2x + 5 = 9$ gives $x = -2$.

3. Domain $\{0,2,4\}$; range $\{-1,3\}$.

4. It suggests fractional tickets are allowed, so separate points are appropriate when only whole tickets can be sold.